

# Elia Gatti

Backend Software Engineer · M.S. Computer Science, University of Trento

elia.gatti01@gmail.com · linkedin.com/in/elia-gatti · github.com/MaiDormo · maidormo.github.io

Available Mar 2027

## Work Experience

### Bitmovin, Software Engineer Intern

June 2026 to Sept 2026

- Shipped KAIROS **end-to-end**, an AI video highlight detection and segmentation service for sports, news and podcasts (VOD plus live): built Next.js frontend flows, a **public API with MCP**, and Go/Python workers backed by Supabase on Terraform-managed GCP. [demo]
- Cut transcription latency by **20%** by tuning worker parallelism and slimming workers, guided by live playback measurements, improving cost and reliability.
- Reduced failed runs and manual operations by building **job orchestration** with startup fixes and **live observability**.
- Backed the API and workers with a **Supabase/PostgreSQL schema**, queries, and **RLS policies**.

### Dedagroup, Software Engineer Intern

May 2024 to Sept 2024

- Halved VM resource requirements by migrating the 'TEN' treasury application from legacy Windows to Linux servers.
- Delivered full-stack features with an HTMX frontend on a Java/Spring backend while maintaining the existing codebase.

## Education

### M.S. in Computer Science, University of Trento

Expected Mar 2027

Thesis (in progress): GPU kernel optimization. GPU programming and high-performance computing.

### B.S. in Computer Science, University of Trento

2024

Bachelor's thesis: MPEG-DASH performance analysis under simulated (Mininet) and real (AWS) networks.

## Projects

### Distributed Key-Value Storage System [code]

- Built a fault-tolerant store with **Akka actors** (Java 21) using consistent hashing and quorum reads/writes, surviving concurrent access and node failures without client disruption.

### MPEG-DASH Performance Analysis (B.S. thesis) [code]

- Measured **adaptive streaming quality** under simulated (Mininet SDN) and real (AWS) networks with Node.js testbed clients exporting playback metrics.

### Sparse Matrix-Vector Multiplication on GPU [code]

- Developed a **hybrid adaptive CUDA kernel** for NVIDIA A30 and profiled execution time and GFLOPS (coalescing, occupancy).

### Guardian: privacy-first eldercare monitor [code] (EuroTech HK Hackathon, Munich, June 2026)

- Built zero-config elderly monitoring with mmWave radar plus an on-device Gemma 4 LLM (no cameras, wearables, or cloud) with behavioral drift tracking and real-time crisis alerting.

## Achievements

### CTM Challenge, Trento 2025: 1st place [code]

- Engineered SVD-based watermark embedding within DWT blocks, robust to compression, noise and transforms; validated with an automated attack suite using ROC analysis.

## Skills

**Languages:** C, C++, Java, Python, Go

**Tools & Frameworks:** PostgreSQL, Supabase, Akka, Docker, Git, Linux, GCP, Terraform, FastAPI, OpenMP, MPI

**Spoken languages:** Italian (native), English (fluent)